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Dynamical Systems - MA3081

Solutions to Exercise Sheet 9

Exercise 9.1 (Center manifolds for maps).

a) In analogy to equation (10.5) in the lecture notes, find the center manifold invariance equation for a discrete-time dynamical system induced by iteration of some map f. You may assume a splitting of f similar to (10.3).

b) Use a) to determine the stability of the origin for the map

$$f(x,y) = \begin{pmatrix} x + xy \\ \frac{1}{2}y - x^2 - xy \end{pmatrix} .$$

Solution.

a) Let us suppose that

$$f : \mathbb{R}^n \to \mathbb{R}^n, \quad (x,y) \mapsto (f_1(x,y), f_2(x,y))$$

has analogously to (10.3) the form

$$f(x,y) = \begin{pmatrix} Bx + F(x,y) \\ Cy + G(x,y) \end{pmatrix} ,$$

where x gives the center directions (i.e. B only has eigenvalues with absolute value 1) and y represents the stable and unstable directions (C has only eigenvalues with absolute value different from 1).

Assume that the center manifold given as the graph of some function h over the variable x and consider some point (x,y) on the manifold, i.e. we have $h(x) = y$. Due to the invariance of the manifold the image $f(x,y) = (f_1(x,y), f_2(x,y))$ of this point (x,y) shall also lie on the graph. Hence we need $h(f_1(x,y)) = f_2(x,y)$ and together with $y = h(x)$ this gives

$$h(f_1(x, h(x))) = f_2(x, h(x)).$$

or using the detailed form above

$$h(Bx + F(x, h(x))) - Ch(x) - G(x, h(x)) = 0 .$$

b) The linearization at the origin is $Df(0,0) = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$, so f is already in the decoupled form. The center space is spanned by $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, i.e. it lies in x-direction, and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ spans the stable eigenspace (y-direction).

We make an ansatz $y = h(x) = ax^2 + \mathcal{O}(x^3)$ for the center manifold, since we know that $h(0) = 0$ and $\mathrm{D}h(0) = 0$. Inserting this into the invariance equation

$$h(x + xh(x)) - \frac{1}{2}h(x) + x^2 + xh(x) = 0$$

and comparing coefficients yields $a = -2$. Hence, the reduced system/map on the center manifold reads

$$\tilde{f} : x \mapsto x + x(-2x^2 + \mathcal{O}\big(x^3\big)) = x - 2x^3 + \mathcal{O}\big(x^4\big) \ . \tag{9.1}$$

Note that for all x sufficiently close to 0 we have $|\tilde{f}(x)| < |x|$, hence the origin is a stable fixed point of (9.1).

Since the non-center-direction y is also stable, the fixed point $(x, y) = (0, 0)$ of f is consequently stable.

□

Exercise 9.2 (Homoclinic bifurcation)**.**

We consider the system

$$\begin{aligned} x' &= -x + 2y + x^2 \,, \\ y' &= (2 - \alpha)x - y - 3x^2 + \frac{3}{2}xy \end{aligned} \tag{9.2}$$

with a parameter $\alpha \in \mathbb{R}$.

a) Determine the eigenvalues and eigenspaces of the linearization at the origin for $\alpha = 0$.

b) Show that for $\alpha = 0$ the equation $x^2(1 - x) - y^2 = 0$ for $x \geq 0$ describes a homoclinic orbit of the system (9.2).

c) Suppose we have a split function $\xi(\alpha)$ with $\xi'(0) < 0$ for a section on the x-axis around $x = 1$ with the same orientation as the axis, i.e. the unstable manifold breaks 'inwards' when we increase α.

   Show that under variation of the parameter α a limit cycle bifurcates from the homoclinic orbit. When does it exist, for $\alpha > 0$ or $\alpha < 0$? Sketch the phase portraits for the three cases $\alpha < 0$, $\alpha = 0$ and $\alpha > 0$.

Solution.

a) For $\alpha = 0$ the vector field is given by

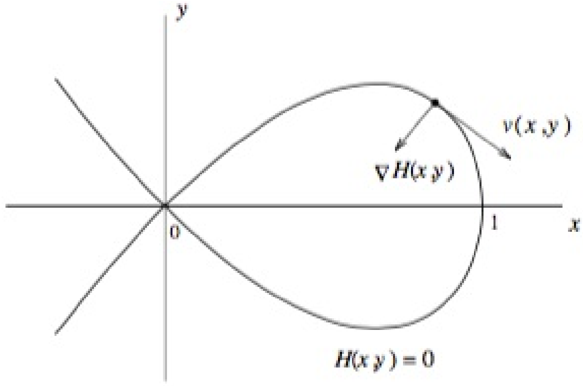
$$f(x, y) = \begin{pmatrix} -x + 2y + x^2 \\ 2x - y - 3x^2 + \frac{3}{2}xy \end{pmatrix} .$$

The Jacobian of f at the origin is

$$Jf(0, 0) = \begin{pmatrix} -1 & 2 \\ 2 & -1 \end{pmatrix}$$

which has eigenvalues $\lambda_1 = -3$, $\lambda_2 = 1$ and corresponding eigenvectors $v_1 = (1, -1)^\top$, $v_2 = (1, 1)^\top$. Hence the origin is a hyperbolic saddle point.

b) Let us first verify that the equation $H(x,y) := x^2(1-x) - y^2 = 0$ defines an orbit of the system. We have to prove that the vector field $f(x,y)$ is tangent to the curve $H(x,y) = 0$ at all points (besides equilibria). Equivalently we can show that $f(x,y)$ is orthogonal along the curve to the normal vector to the curve. Note that a normal vector to the level set $\{H(x,y) = 0\}$ is given by the gradient of H.



Hence we compute

$$\begin{aligned}
\langle f(x,y), (\nabla H)(x,y)\rangle &= (-x+2y+x^2)(2x-3x^2) + (2x - y - 3x^2 + \tfrac{3}{2}xy)(-2y) \\
&= -2x^2 + 4xy + 2x^3 + 3x^3 - 6x^2y - 3x^4 - 4xy + 2y^2 + 6x^2y - 3xy^2 \\
&= -2x^2 + 2x^3 + 3x^3 - 3x^4 + 2y^2 - 3xy^2 \\
&\overset{(*)}{=} -2x^2 + 2x^3 + 3x^3 - 3x^4 + 2[x^2(1-x)] - 3x[x^2(1-x)] \\
&= -2x^2 + 2x^3 + 3x^3 - 3x^4 + 2x^2 - 2x^3 - 3x^3 + 3x^4 \\
&= 0
\end{aligned}$$

where we have used at $(*)$ that we are on the curve $H(x,y) = 0$, so $y^2 = x^2(1-x)$.

Next, we observe that we have an equilibrium point at the cusp of the curve at the origin. It remains to show that there are no further equilibria on the curve so that the whole orbit is run through and the origin is reached in forward and backward time.

Taking $y = \pm x\sqrt{(1-x)}$ and plugging this into the first differential equation yields $x' > 0$ for all $y > 0$ and $x' < 0$ for all $y < 0$, so there cannot be an equilibrium on the curve whenever $y \neq 0$. Furthermore, at the remaining point $(x,y) = (1,0)$ we have $y' = -1$, hence $(x,y) = (0,0)$ is the only equilibrium on the curve. Consequently it describes a homoclinic orbit.

c) We apply the Andronov-Leontovich Theorem. From a) we know that

$$\sigma(0) := \lambda_1(0) + \lambda_2(0) = -3 + 1 = -2 \neq 0.$$

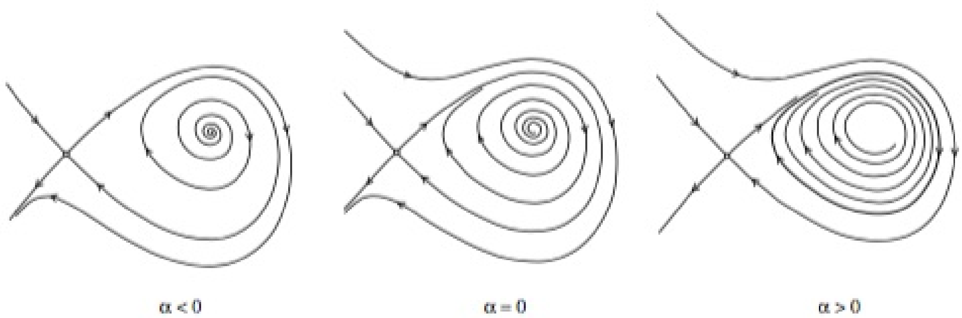
Since by assumption also the "breaking" condition of Theorem 12.3 is fulfilled, we can conclude that for $|\alpha|$ sufficiently small a unique limit cycle bifurcates from the homoclinic orbit. Since $\sigma(0) < 0$, we know that the limit cycle is stable. Then, the stability of the limit cycle allows to conclude for which parameter values the cycle exists: Assuming that we are in the situation of a limit cycle, the "inner" manifold (in our case W^s for $\alpha < 0$ and W^u for $\alpha > 0$) is trapped between the cycle and the other manifold. Hence it must converge to the limit cycle in forward time (in case of W^u) or backward time (in case of

W^s). Consequently, a stable limit cycle is only possible if W^u is the "inner" manifold (in case that W^s is the "inner" manifold we obtain an unstable limit cycle).

Alternatively, one can deduce from the proof of Theorem 12.3 that the limit cycle exists for $\sigma(0) < 0$ and $q > 0$. The only complication is that, in the proof, we consider a section that is oriented "inwards", while in this exercise we are given a section that is oriented "outwards", and hence $q > 0$ (in the proof) corresponds to $\xi < 0$ (here).

From the sign of the split function we know that in our case the unstable manifold breaks inward for $\alpha > 0$ and outward for $\alpha < 0$. Thus the limit cycle occurs for $\alpha > 0$.

The phase portraits look as follows:



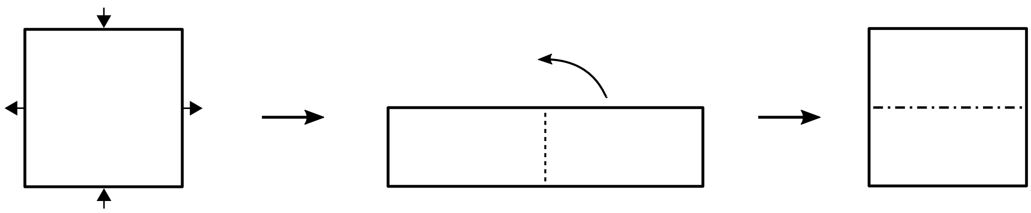
□

Exercise 9.3 (Baker's map)**.**

On the unit square we consider the baker's map

$$g\colon [0,1)^2 \to [0,1)^2\,, \qquad g(x,y) = \begin{cases} \left(2x,\ \tfrac{1}{2}y\right) & \text{if } x \in [0,\tfrac{1}{2})\,, \\ \left(2x-1,\ \tfrac{1}{2}y + \tfrac{1}{2}\right) & \text{if } x \in [\tfrac{1}{2},1)\,. \end{cases}$$

This map represents a baker handling dough. He rolls out the dough to twice the width but half the height. The flat dough is then cut into two pieces and the right half is placed on top of the other.



For the two sets $A = [0,\frac{1}{2})\times[0,1)$ and $B = [\frac{1}{2},1)\times[0,1)$ determine the images $g(S), g^2(S), g^3(S), \ldots$ and preimages $g^{-1}(S), g^{-2}(S), g^{-3}(S), \ldots$ with $S \in \{A, B\}$.

Next, take some initial condition $(x,y) \in [0,1)^2$ and write it in its binary representation. What can you observe when you apply the map g?

Solution. The first three images and preimages of A and B are illustrated in Figure 1. They respectively consist of horizontal/vertical strips of width 2^{-n}.

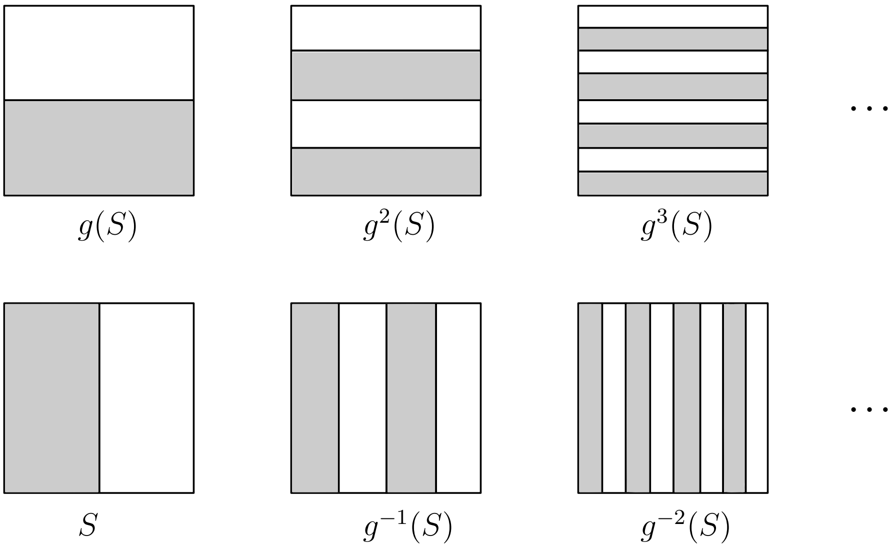


Figure 1: Images and preimages of the sets A (grey) and B (white) under iterations of g.

Next, let us take some initial condition $(x, y) \in [0, 1)^2$ with binary representations

$$\begin{aligned} x &= 0, x_1 x_2 x_3 ... = \sum_{k=1}^{\infty} x_k \, 2^{-k} \\ y &= 0, y_1 y_2 y_3 ... = \sum_{k=1}^{\infty} y_k \, 2^{-k} \end{aligned}$$

with digits/coefficients $x_k, y_k \in \{0, 1\}$ for all $k \in \mathbb{N}$.
We observe that

$$x_1 = 0 \text{ if } (x, y) \in A \text{ and } x_1 = 1 \text{ if } (x, y) \in B\,.$$

Furthermore, we find that

$$x_k = 0 \text{ if } (x, y) \in g^{-k+1}(A) \text{ and } x_k = 1 \text{ if } (x, y) \in g^{-k+1}(B)\,.$$

For the y coordinate we obtain similar results with forward images, i.e.

$$y_k = 0 \text{ if } (x, y) \in g^{k}(A) \text{ and } y_k = 1 \text{ if } (x, y) \in g^{k}(B)\,.$$

Let us now have a closer look what happens with the binary representations if we apply the map

$$g\colon [0,1)^2 \to [0,1)^2\,, \qquad g(x,y) = \begin{cases} \left(2x,\ \frac{1}{2}y\right) & \text{if } x \in [0, \frac{1}{2})\,, \\ \left(2x-1,\ \frac{1}{2}y + \frac{1}{2}\right) & \text{if } x \in [\frac{1}{2}, 1)\,. \end{cases}$$

In the first component we have

$$\begin{aligned} g_1(x,y) &= 2x \pmod 1 = 2 \cdot \sum_{k=1}^{\infty} x_k \, 2^{-k} \pmod 1 = \sum_{k=1}^{\infty} x_k \, 2^{-k+1} \pmod 1 = \sum_{k=0}^{\infty} x_{k+1} \, 2^{-k} \pmod 1 \\ &= x_1, x_2 x_3 ... \pmod 1 = 0, x_2 x_3 ... \end{aligned}$$

so the binary representation is shifted by one position to the left (and the first digit is deleted).

For the second component we obtain in the case of $x \in [0, \frac{1}{2})$

$$g_2(x,y) = \tfrac{1}{2}y = \tfrac{1}{2} \cdot \sum_{k=1}^{\infty} y_k \, 2^{-k} = \sum_{k=1}^{\infty} y_k \, 2^{-k-1} = \sum_{k=2}^{\infty} y_{k-1} \, 2^{-k} = 0,0y_1y_2y_3...$$

and if $x \in [\frac{1}{2}, 1)$

$$g_2(x,y) = \tfrac{1}{2}y + \tfrac{1}{2} = \tfrac{1}{2} \cdot \sum_{k=1}^{\infty} y_k \, 2^{-k} \ + \tfrac{1}{2} = \sum_{k=1}^{\infty} y_k \, 2^{-k-1} \ + \tfrac{1}{2} = \sum_{k=2}^{\infty} y_{k-1} \, 2^{-k} \ + \tfrac{1}{2} = 0,1y_1y_2y_3...$$

Hence the binary representation of y gets shifted by one position to the right and depending on x a 0 or 1 is inserted at the first position after the comma. Since x_1 determines if $x < \frac{1}{2}$ or $x \geq \frac{1}{2}$ we can summarize this to

$$g_2(x,y) = 0, x_1y_1y_2y_3... \, .$$

Let us now put the binary coefficients (after the comma) of x and y together so that they form a bi-infinite sequence

$$\ldots y_3 \, y_2 \, y_1 \, x_1 \, x_2 \, x_3 \ldots$$

Note that the first digit of x, that was removed when applying g, reappears in the first position of the second component $g_2(x,y)$. Thus the induced action of the map g on this bi-infinite sequence is just a left shift. Due to this relation to the left shift operator we can also expect chaotic behaviour for the baker's map.

$\square$